

INTRODUCTION TO MACHINE LEARNING

Basic Concepts, Supervised Learning, Unsupervised Learning

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2026-III



Outline

- 1 Fundamentals of Machine Learning ✓
- 2 Supervised Machine Learning ✓
- 3 Unsupervised Machine Learning ✓

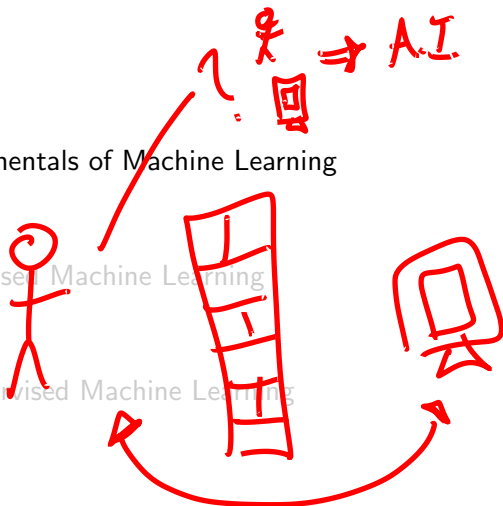


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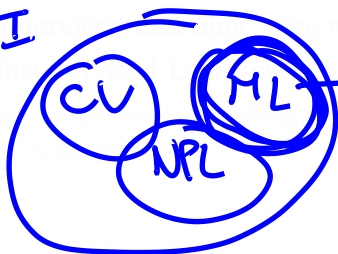
Key Concepts in Machine Learning

 $10^3 \rightarrow 10^6$

Machine Learning

- **Machine learning** is a method of data analysis that **automates** analytical model building.
- It is a **branch** of **artificial intelligence** based on the idea that systems can **learn from data**, **identify patterns** and **make decisions** with minimal human intervention.

AI



data for learning

big volume

\$ 2000 business B.I.

GPT-4.5 $\Rightarrow$ 1 trillion) 250M USD 4 weeks

Key Concepts in Machine Learning

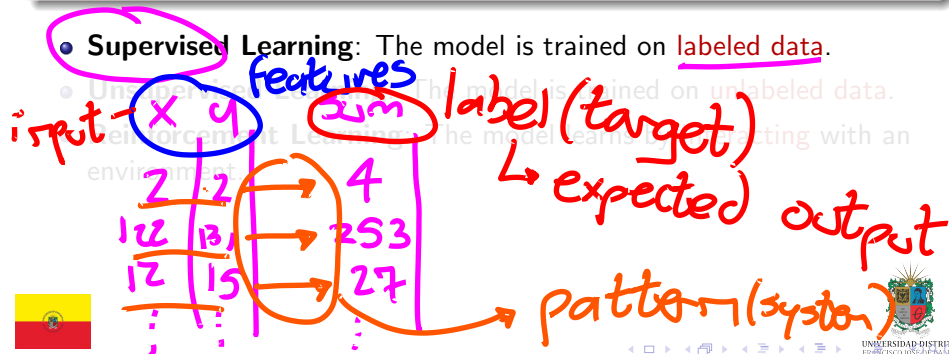
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- **Supervised Learning:** The model is trained on labeled data.

- **Unsupervised Learning:** The model is trained on unlabeled data.

- **Reinforcement Learning:** The model learns by interacting with an environment.

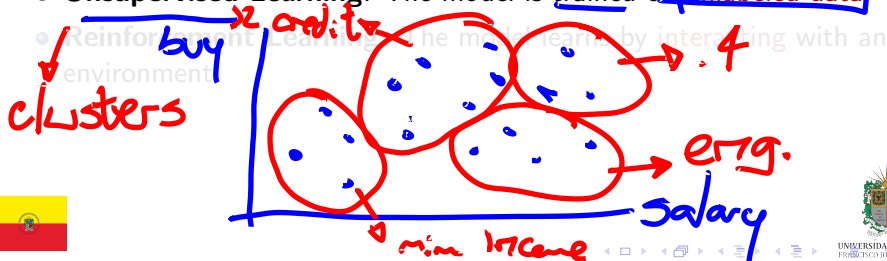


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input → output
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Typical Machine Learning Problems

- **Classification:** Predicting a **label**.
- **Regression:** Predicting a continuous value.
- **Clustering:** Grouping similar data points.
- **Dimensionality Reduction:** Reducing the number of features.
- **Anomaly Detection:** Identifying unusual data points.
- **Association Rule Learning:** Identifying relationships between variables.



cluster
damage

category
supervised
unsupervised

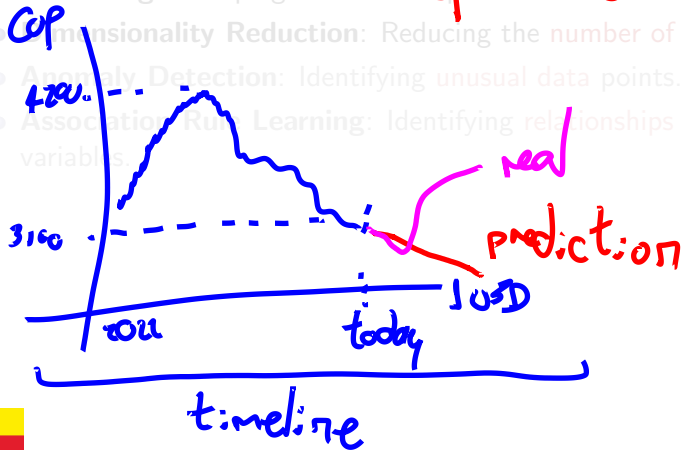


Typical Machine Learning Problems

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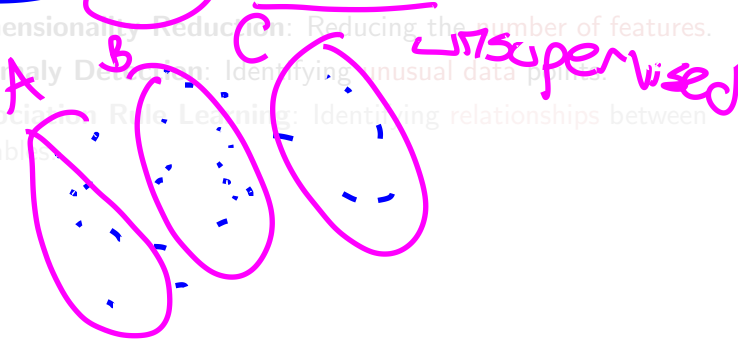
- **Clustering:** Grouping similar **unlabeled** data points.
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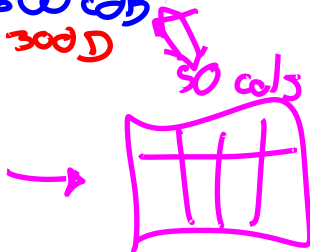
• **Anomaly Detection:** Identifying **unusual** data points.

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Handwritten notes above the table:

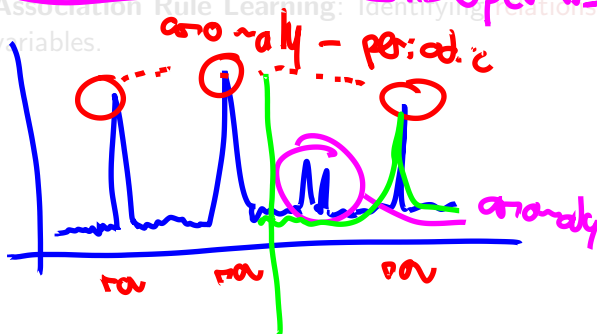
- 100D (in red)
- 200 cols - 300 cols (in blue)
- Superv. sev (in blue)

-	-	-	-	-	-



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- **Association Rule Learning:** Identifying relationships between variables. *anomaly - periodic*



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pre
processing

A

•

•

•

B

•

•

•

morning

↳ sun

↳ wind

rain

afternoon



The Machine Learning Workflow

- **Data Collection:** Gathering the data.

- **Data Preprocessing:** Cleaning and preparing the data.

- **Feature Engineering:** Creating new features.

- **Model Selection:** Choosing the best model.

- **Model Training:** Training the model on the data.

- **Model Evaluation:** Assessing the model's performance.

- **Model Deployment:** Putting the model into production.

data sources + ETL + database

↓
data warehouse

data engineers



The Machine Learning Workflow

- **Data Collection:** Gathering the **data**.
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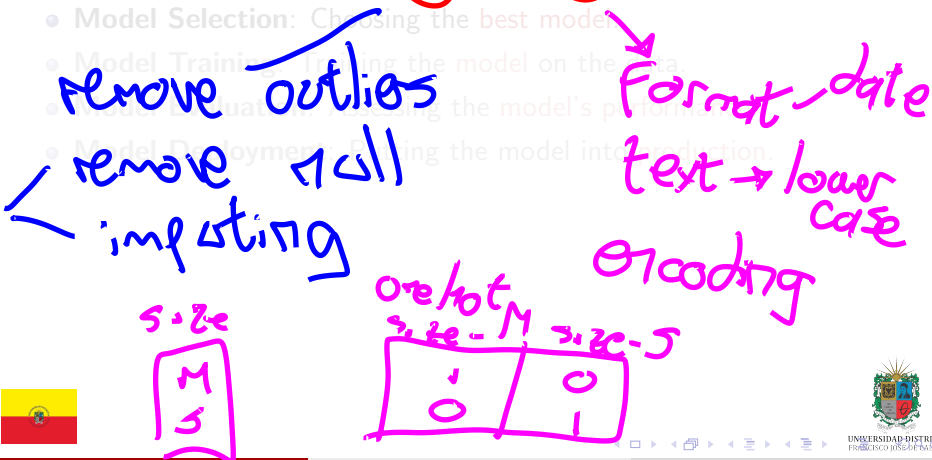
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X

train	val	test
4.5	2.0	
4.1	4.4	
3.0	3.0	

decr.
↓
0
0



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EDA

SUP-

- class

KNN

ANN

- reg

x Boost

;

SUP-

- class

;

- clust.

;



The Machine Learning Workflow

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Handwritten notes illustrating the workflow:

Model Evaluation: Assessing the model's performance.

Model Deployment: Fitting the model into production.

Handwritten calculations (likely representing feature engineering or model output):

- $2 \times 2 = 4$
- $2 \times 3 = 6$
- $3 \times 2 = 6$
- $3 \times 3 = 9$
- $4 \times 2 = 8$
- $4 \times 3 = 12$
- $2 \times 9 = 18$
- $3 \times 9 = 27$
- $4 \times 9 = 36$

Handwritten text: **learn** → **pattern detection**



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$$2 \times 3 = ? \quad (6)$$

$$4 \times 2 = ? \quad (8)$$

$$4 \times 3 = ? \quad (12)$$

$$3 \times 9 = ? \quad (27)$$

learn $\rightarrow$ less error



The Machine Learning Workflow

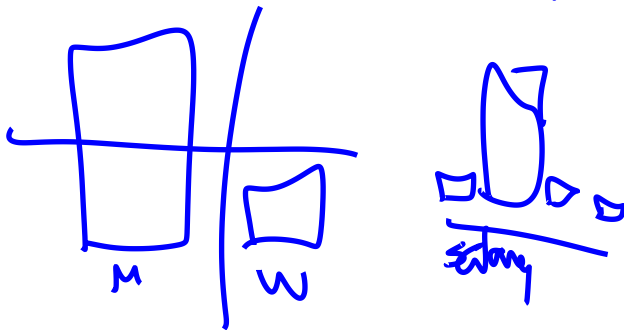
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Algorithmic Bias — sesgo

- **Algorithmic bias** is a **systematic error** in a model that results in **unfair outcomes**.
- It can be caused by **biased training data**, biased algorithms, or biased decision-making.

→ avoid → pre-processing



Outline

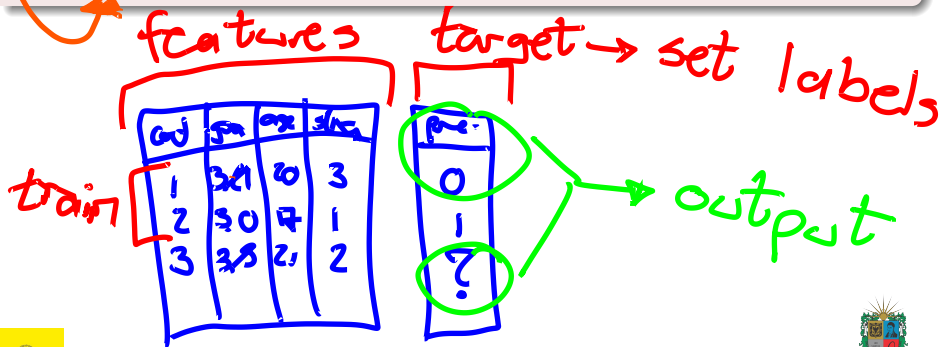
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Introduction to Supervised Machine Learning

Definition

- **Supervised learning** is a type of machine learning where the model is trained on labeled data.
- It involves training a model to map input data to output data based on example input-output pairs.



Overfitting and Underfitting

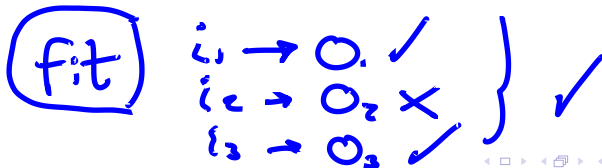
Overfitting

Overfitting occurs when a model learns the training data too well and performs poorly on new data.

Underfitting

Underfitting occurs when a model is too simple to capture the underlying structure of the data.

train ✓ test x



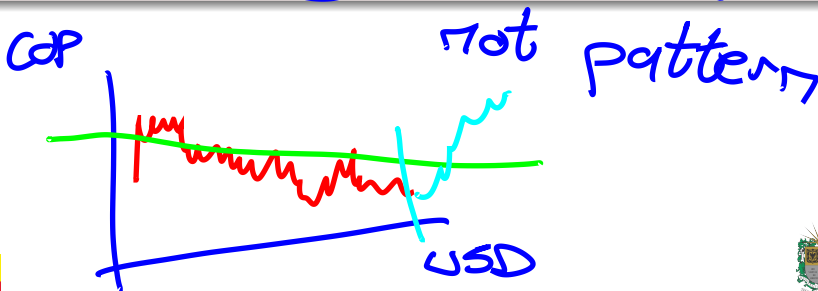
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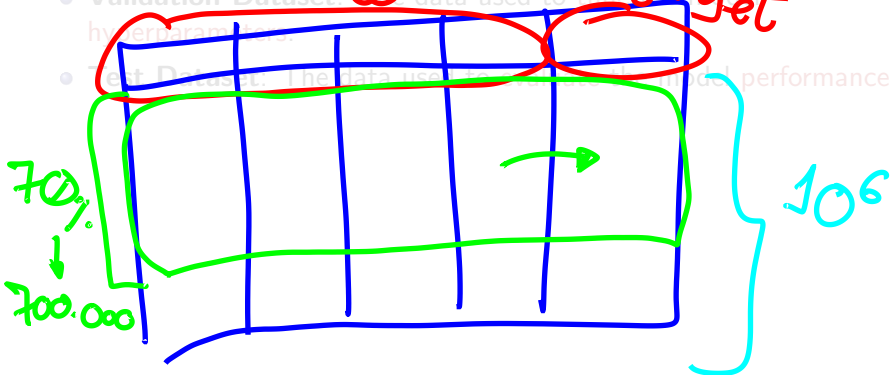
Underfitting occurs when a model is too simple to capture the underlying structure of the data.



Supervised Learning Datasets

- **Training Dataset:** The data used to train the model.

- **Validation Dataset:** The data used to tune the model's hyperparameters.
- **Test Dataset:** The data used to evaluate the model's performance.



learn → less errors

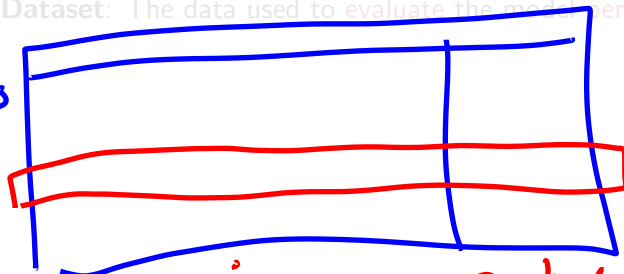


Supervised Learning Datasets

- **Training Dataset:** The data used to **train the model**.
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train
in
libraries

10%
100.000



hyperparameter
tuning

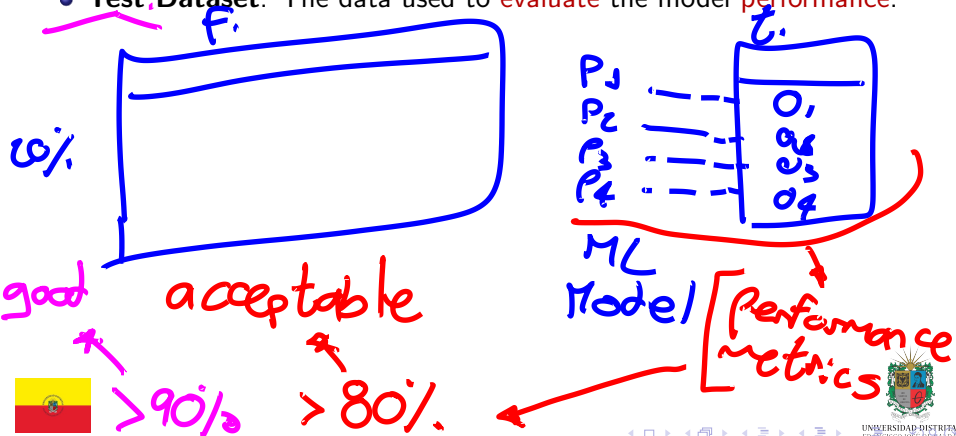
prediction target
error



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

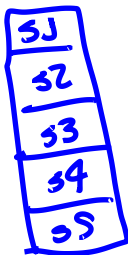
Supervised Learning Datasets

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Cross-Validation

- **Cross-validation** is a technique for **assessing the performance** of a model.
- It involves **splitting** the **data** into multiple **subsets**, training the model on some subsets, and evaluating it on others.
- Common cross-validation **techniques** include **k-fold cross-validation** and **leave-one-out cross-validation**.
- Cross-validation helps to **reduce overfitting** and provides a more **accurate estimate** of the model's **performance**.



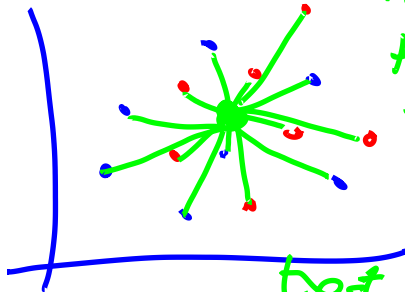
train: s1 s2 s3 s4 test: s5
 train: s2 s3 s4 s5 test: s1
 train: s3 s4 s5 s1 test: s2
 " s4 s5 s1 s2 s3
 " s5 s1 s2 s3



K-Nearest Neighbors

- **K-Nearest Neighbors (KNN)** is a simple algorithm that stores all available cases and classifies new cases based on a **similarity measure**.
- It can be used for both **classification** and **regression** tasks.
- For **classification**, the output is the **class label** of the majority of the k-nearest neighbors.

$x \quad y \quad l$
 $c_1, c_2 \quad r$
 $c_3, c_4 \quad b$
 $\vdots \quad \vdots$
 $\vdots \quad \vdots$



$k=1 \rightarrow b$

$k=3 \rightarrow r$

$k=5 \rightarrow r$

$\vdots$
~~best~~
 best - r.



fit history

- **Linear regression** is a type of regression analysis used for predicting the value of a continuous dependent variable.
- It works by finding the line that best fits the data.

Least Squares

~~Least squares is a method for finding the best fitting line by minimizing the sum of the squared differences between the predicted and actual values.~~

$$f(x) = \underline{ax} + \underline{b}$$

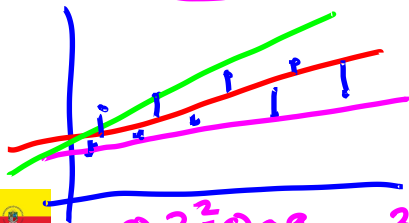
Linear Regression with Least Squares

Linear Regression

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Least Squares

Least squares is a method for finding the **best-fitting line** by **minimizing the sum** of the squared differences between the **predicted** and **actual** values.

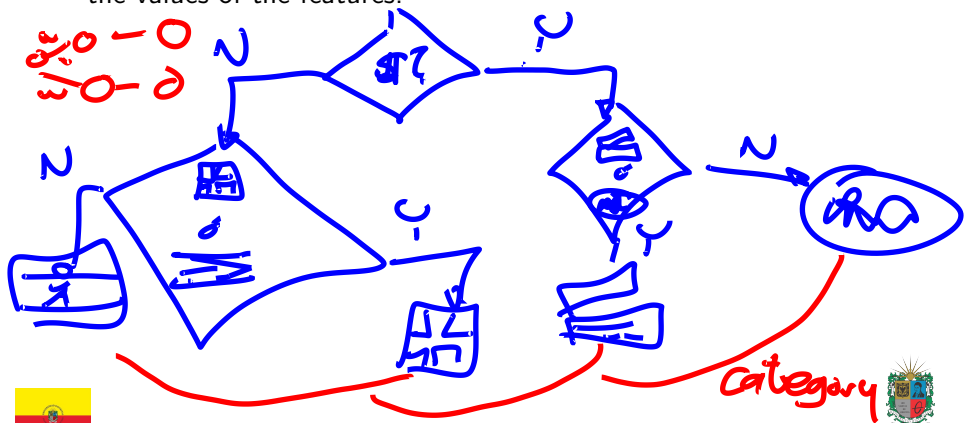


min distance $\sum$
error



Decision Trees

- **Decision trees** are a type of **machine learning model** that can be used for both **classification** and **regression** tasks.
- They work by recursively **partitioning** the **data** into **subsets** based on the values of the features.



Random Forest

100 trees

- **Random forest** is an **ensemble learning** method that combines **multiple decision trees** to create a strong predictive model.
- It works by building **multiple trees** and **averaging** their predictions to **reduce overfitting**.



less error



Neural Networks

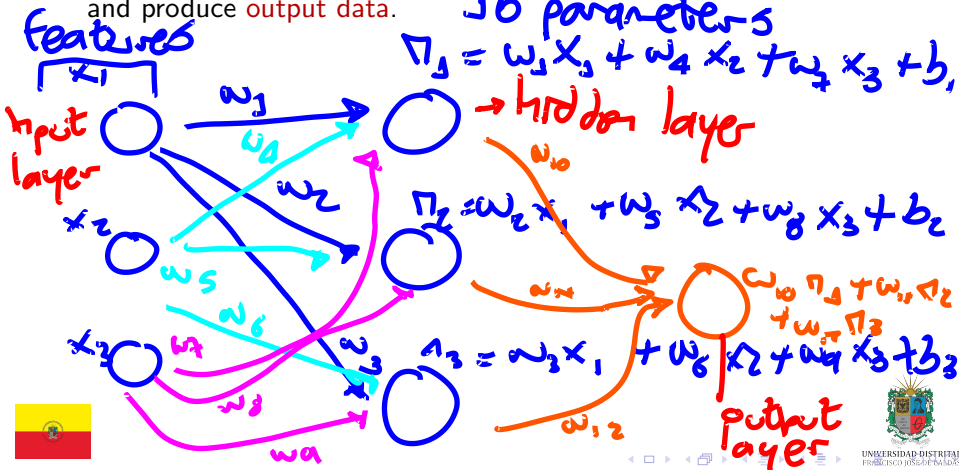
~ 1943 / Perceptron / Backprop
 ~ 50 / 70

- **Neural networks** are a type of machine learning model inspired by the human brain.

$$f(x) = ax + b$$

- They consist of layers of interconnected nodes that process input data and produce output data.

16 parameters



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Introduction to Unsupervised Machine Learning

Definition

- **Unsupervised learning** is a type of **machine learning** where the model is trained on **unlabeled data**.
- It involves finding **patterns** and **relationships** in the data without any predefined labels.



Clustering

- **Clustering** is a type of **unsupervised learning** that involves **grouping** similar **data points** together.
- Common clustering algorithms include **k-means**, **hierarchical clustering**, and **DBSCAN**.



K-means

- **K-means** is a popular **clustering algorithm** that partitions the data into K **distinct clusters** based on feature similarity.
- It works by iteratively assigning data points to the nearest **cluster centroid** and updating the centroids based on the assigned points.



Anomaly Detection

- **Anomaly detection** is a type of **unsupervised learning** that involves identifying **unusual data points** in a dataset.
- Common anomaly detection algorithms include **Isolation Forest**, **One-Class SVM**, and **Autoencoders**.



Autoencoders

- **Autoencoders** are a type of **neural network** used for unsupervised learning.
- They work by **encoding** the input data into a lower-dimensional representation and then **decoding** it back to the original data.



Dimensionality Reduction

- **Dimensionality reduction** is a technique for **reducing** the number of **features** in a dataset while retaining as much information as possible.
- Common dimensionality reduction techniques include **Principal Component Analysis (PCA)** and **t-Distributed Stochastic Neighbor Embedding (t-SNE)**.



Principal Component Analysis (PCA)

- **Principal Component Analysis (PCA)** is a **statistical technique** used for **dimensionality reduction**.
- It works by transforming the data into a **new coordinate system** where the greatest variance lies along the first principal **component**, the second greatest variance along the second principal **component**, and so on.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/machine-learning>

